

Interaction Transfer

Toward a Theory of Experience and Adaptation

Cross-disciplinary research paper

Abstract

Interaction transfer is the influence of prior interaction experience on performance, interpretation, opportunity recognition, strategy, or learning in a later interaction under changed conditions. It is not synonymous with learning, familiarity, repetition, expertise, or successful generalization. Learning is change attributable to experience; transfer is evidence that some consequence of prior experience affects a distinct target context. Familiarity denotes prior exposure and may produce fluency without transferable structure. Expertise is durable, organized performance within a domain and may remain highly context-specific.

Across cognitive psychology, learning sciences, skill acquisition, expertise research, reinforcement learning, transfer learning, activity theory, distributed cognition, and HCI, existing theories provide the required mechanisms: common elements, abstraction, analogical mapping, schema induction, procedural compilation, policy reuse, representation alignment, retrieval, contextual cueing, habit, metacognitive control, and socially distributed practice. No new mental substance called interaction memory is required. What is missing is a shared interaction-level contract that specifies the source episode, target episode, transferred content, mediating mechanism, preserved relations, context shift, measurement model, and counterfactual evidence.

The paper therefore concludes that existing theories are sufficient but require a typed cross-disciplinary transfer framework. The proposed model represents experience as changes to agent, artifact, environment, and coordination state. Transfer is classified by what changes, where it is expressed, its direction, distance, durability, intentionality, and dependence on representation. The reusable artifacts include a transfer ontology, adaptation model, transfer taxonomy, interaction-experience model, metric registry, transfer contract, comparison protocol, and falsification matrix.

1. Problem Statement

Prior interaction often changes later interaction, but the phrase “experience transfers” compresses several different claims. A person may become faster because the same motor sequence is repeated, may recognize a structurally similar command in a new interface, may reuse a strategy across software families, may apply an inappropriate habit, or may reconstruct a solution through analogy. An artificial agent may initialize a policy from a related task, reuse a representation, or exploit shared dynamics. A team may retain coordination routines in artifacts and procedures rather than in any single member.

A scientific theory must therefore answer four questions separately: what changed because of the source interaction; what component of that change was available in the target; what relation between source and target licensed reuse; and what evidence shows that target behavior depended on the source rather than on baseline ability, repeated exposure, task similarity, or measurement artifact.

2. Method and Scope

The review used concept comparison rather than disciplinary summary. Foundational accounts of transfer, skill, expertise, analogy, situated learning, reinforcement learning, transfer learning, distributed cognition, and HCI migration were compared by their units, mechanisms, boundary conditions, and evidence requirements. The analysis treated the completed Interaction, Behavior, Measurement, Opportunity, Representation, and Similarity frameworks as constraints, not as evidence that a new construct is needed.

The main falsification question was whether “interaction transfer” adds explanatory content beyond existing theories. The candidate framework was retained only where it coordinates distinctions that established theories already support but do not jointly expose.

3. Existing Theories of Transfer

3.1 Common elements and structural overlap

Thorndike and Woodworth’s identical-elements tradition explains transfer through overlap between source and target demands. Later cognitive accounts replace literal stimulus identity with shared productions, procedures, relations, or representational components. This family explains why near transfer is common and why far transfer is difficult. Its limit

is that “shared element” depends on the level of representation: two interfaces can differ visually while preserving an operation grammar, or appear similar while implementing different semantics.

3.2 Abstraction, schema, and analogy

Schema and analogical-transfer theories explain reuse when a learner abstracts relational structure from examples, retrieves a relevant source, maps corresponding relations, and adapts the solution. Transfer may fail at retrieval even when a useful analogy exists, or fail at mapping when superficial similarity dominates. Representation therefore affects transfer by changing which invariants are salient and recoverable, not by directly determining behavior.

3.3 Skill acquisition and procedural knowledge

Skill theories distinguish declarative knowledge, proceduralization, compilation, chunking, automaticity, and adaptive control. Practice can reduce time and error while narrowing flexibility. Procedural knowledge transfers when target conditions preserve relevant productions or control structures; automatic habits can create negative transfer when changed contingencies are not detected.

3.4 Expertise

Expertise reorganizes perception, memory, diagnosis, and strategy around domain-relevant structure. Experts notice relations that novices miss, but expert performance is not a general transferable capacity. It is often conditional on representative tasks, feedback histories, and domain-specific knowledge. Expertise must therefore be measured as a profile of calibrated performance and representation, not inferred from tenure or status.

3.5 Reinforcement learning and transfer learning

Reinforcement learning formalizes transfer through reused value functions, policies, representations, options, models, demonstrations, or successor features. Machine-learning transfer theory separates source and target domains and tasks, and recognizes negative transfer when reused structure harms target performance. These formalisms are precise about optimization but often abstract away human interpretation, cultural practice, artifact mediation, and discoverability.

3.6 Situated, activity, and distributed accounts

Situated and activity theories challenge the assumption that transferable knowledge is an internal package carried unchanged between contexts. What persists may be participation, routines, artifact configurations, conventions, or access to distributed resources. Transfer can be a transformation of participation or coordination rather than retrieval of an invariant internal representation. A general framework must therefore permit agent-internal, artifact-mediated, social, and environmental carriers.

4. Canonical Distinctions

Learning is a relatively durable change in capability, representation, policy, or behavior attributable to experience. Transfer is a source-dependent difference in a target context that is not identical to the learning context. Adaptation is change that improves or regulates fit to current conditions; it may arise without transfer from a distinct source. Familiarity is prior exposure to a person, representation, operation, or context; it is a predictor, not a mechanism or outcome. Practice is repeated performance. Expertise is organized, reliable, and often efficient performance within a domain. Habit is cue-dependent action tendency strengthened by repetition. Generalization is extension of a learned function to unobserved cases and is one formal species of transfer. Retention concerns persistence over time without requiring contextual change.

5. What Transfers?

The literature does not support one transferable entity. At least eight carrier types recur:

- Perceptual attunement: sensitivity to cues, invariants, or diagnostic structure.
- Semantic knowledge: concepts, relations, rules, and causal models.
- Procedures and skills: action sequences, productions, motor programs, and operation grammars.
- Strategies and policies: conditional organization of actions under goals and contingencies.
- Expectations and beliefs: anticipated availability, consequence, risk, and reversibility.
- Values and preferences: learned utility, cost, confidence, and avoidance structure.

- Externalized structure: notes, templates, macros, layouts, documentation, and shared artifacts.
- Coordination practices: roles, conventions, communication routines, and division of labor.

These carriers can transfer independently. A user may retain semantic knowledge but lose motor fluency after modality migration; preserve a strategy while relearning commands; retain a habit that becomes harmful; or use an external artifact without internal recall.

6. Interaction Experience Model

An interaction experience is represented as an update, not merely an episode record. Let an experience update be:

$$\Delta E = \langle \Delta K, \Delta P, \Delta B, \Delta V, \Delta H, \Delta A, \Delta C, \Delta M, U \rangle$$

where ΔK is change in semantic or causal knowledge; ΔP is procedural or policy change; ΔB is change in beliefs about actions and consequences; ΔV is value or preference change; ΔH is habit strength; ΔA is perceptual attunement; ΔC is change in coordination practices; ΔM is change to external memory or artifacts; and U records uncertainty and attribution.

The source interaction does not deposit a complete transferable object. It changes a distributed experience state spanning agent, artifacts, environment, and social organization. A later target interaction recruits some subset of that state under target cues and constraints.

7. Candidate Adaptation Model

Transfer from source S to target T is modeled as a sequence:

1. Acquisition: source interaction changes one or more experience-state components.
2. Retention: some change persists until the target episode.
3. Access: target cues, goals, or deliberate search activate relevant prior structure.
4. Correspondence: source and target elements or relations are aligned.
5. Reuse: prior knowledge, procedure, policy, artifact, or coordination routine is applied.
6. Adaptation: reused structure is modified for target constraints.
7. Evaluation: feedback reveals fit, mismatch, or uncertainty.
8. Consolidation or revision: the experience state is strengthened, differentiated, suppressed, or reorganized.

Transfer success is therefore conjunctive. Failure can arise from absence of learning, decay, retrieval failure, poor correspondence, rigid reuse, inadequate opportunity, altered authority, incompatible representation, or misleading feedback.

8. Formal Definition

Let S and T be source and target interaction contexts under declared similarity and measurement contracts. Let Y_T be a target measure and $do(S=1)$ denote exposure to the source experience. Transfer exists for population or agent x when:

$$\tau = E[Y_T | do(S=1)] - E[Y_T | do(S=0)] \neq 0$$

and the target differs from the source on at least one declared contextual or representational dimension. Positive transfer has τ in the desired direction; negative transfer has τ in the harmful direction; null transfer is compatible with learning when source gains do not affect the target.

Mechanistic transfer claims additionally require evidence that a specified carrier or mediator changed after S , was available or retrieved in T , and accounted for part of τ under plausible alternatives. Observed target superiority alone establishes neither the mechanism nor the transferred content.

9. Transfer Taxonomy

Transfer should be typed along independent axes rather than labeled simply near or far:

Axis	Values	Question
Direction	positive, negative, null, mixed	Did prior experience help, harm, or have heterogeneous effects?
Distance	declared multidimensional source-target difference	Across what changes in task, representation, modality, device, domain, time, and social context?
Carrier	knowledge, procedure, policy, belief, value, habit, attunement, artifact, coordination	What consequence of experience was reused?
Access	spontaneous, cued, prompted, instructed	How was prior structure recruited?
Level	event, action, episode, strategy, practice	At what behavioral organization was transfer expressed?
Duration	immediate, delayed, durable, decaying	How long did the effect persist?
Intentionality	deliberate, incidental, automatic	Was transfer consciously attempted?
Distribution	individual, artifact-mediated, collective, environmental	Where was transferable structure carried?
Specificity	token, type, relational, principle-based	What invariant supported correspondence?

10. Comparative Cases

10.1 Interface migration

Migration between software versions can preserve task semantics while changing layout, labels, shortcuts, or workflow. Semantic and strategic transfer may be positive, motor and habitual transfer negative, and discoverability reduced. “Familiar interface” is therefore not a unitary condition.

10.2 Modality migration

Moving from visual to auditory or tactile interaction may preserve operations and outcomes while changing seriality, memory load, cue persistence, navigation, and error recovery. Transfer should be assessed separately for semantic knowledge, strategy, command vocabulary, spatial mapping, and timing.

10.3 Cross-device interaction

Desktop, tablet, and mobile variants may share content and account state but differ in gesture vocabulary, screen real estate, interruption patterns, and exposed opportunity. Transfer depends on preserved operation grammar and representation cues, not brand identity alone.

10.4 Software families

Shared conventions across applications can support rapid transfer of editing, navigation, and file-management procedures. False friends - identical symbols with different effects - produce negative transfer. Family resemblance must be decomposed into semantic, structural, opportunity, and protocol similarity.

10.5 Expert and novice

Experts often transfer abstract relations and diagnostic cues, while novices rely on surface correspondence. Yet expertise can impair transfer when automated routines suppress detection of changed constraints. Group labels should be replaced by measured knowledge, strategy, calibration, and source exposure.

10.6 Repeated task performance

Improvement on the same task establishes practice and retention, not transfer. A valid transfer test must alter at least one declared dimension and preserve a defensible basis for comparison. Otherwise source and target are repetitions of one condition.

11. Measurement and Metrics

Transfer measurement must inherit the Measurement Contract. The source and target constructs, observation models, scales, segmentation, uncertainty, and comparison frame must be declared. Recommended metrics include:

- Target effect: controlled difference in target performance attributable to source exposure.
- Transfer efficiency: target gain relative to source practice cost or exposure.
- Savings: reduction in trials, time, or error needed to reach a criterion.
- Generalization gradient: effect magnitude across declared source-target distances.
- Belief transfer: change in precision, recall, and calibration of believed action sets.
- Strategy reuse: probability of an inferred strategy under target contingencies, with uncertainty.
- Negative-transfer cost: excess error, latency, recovery, or risk caused by inappropriate reuse.
- Durability: transfer effect as a function of delay.
- Adaptation rate: speed with which reused structure is revised after mismatch.
- Representation sensitivity: interaction between source-target representation change and transfer.

Self-report of usefulness is evidence about perceived transfer, not proof of behavioral transfer. Faster target performance may reflect familiarity, motor priming, motivation, selection, or common task ability. Mediation claims require temporally ordered carrier measurements and sensitivity analysis.

12. Transfer Protocol

1. Declare source and target interaction contracts.
2. Specify which target dimensions differ and which invariants license comparison.
3. Define the candidate carrier and mechanism before observing target outcomes.
4. Measure source learning separately from target transfer.
5. Include a no-source or alternative-source counterfactual.
6. Control prior exposure, baseline ability, motivation, device access, and opportunity structure.
7. Test immediate and delayed target performance where durability matters.
8. Measure positive, negative, and null outcomes, including recovery and strategy change.
9. Separate direct observations, derived features, latent carriers, and causal conclusions.
10. Report uncertainty, heterogeneity, exclusions, and boundary conditions.
11. Replicate across at least one additional target context before claiming generality.

13. Falsification Matrix

Claim	Evidence against	Consequence
Interaction transfer needs a special primitive	Established learning and transfer models explain all observed effects with no loss	Reject new primitive; retain only coordination vocabulary
Source-target similarity predicts transfer	Transfer is unrelated to declared similarities after controls	Revise similarity dimensions or reject overlap mechanism
Belief mediates interface transfer	Belief change is unreliable or does not predict target action	Remove belief mediation claim
Strategy reuse supports far transfer	Target gains occur without evidence of strategy access or adaptation	Treat outcome gain as unexplained transfer
Distributed carriers matter	Removing artifacts and coordination resources leaves effects unchanged	Narrow framework to agent-internal carriers for that domain
Transfer differs from familiarity	Exposure-only controls explain the full target effect	Classify effect as familiarity or repetition, not structural transfer
Framework improves prediction	Typed variables add no held-out gain over source exposure and target similarity	Reduce framework to descriptive taxonomy

14. Threats to Validity

Transfer is unusually sensitive to source-target definition. Researchers can manufacture “far transfer” by describing contexts broadly or manufacture “near transfer” by choosing fine-grained dimensions. Distance must therefore be a declared vector, not an intuitive label.

Selection and expectancy threaten causal inference: experienced users differ before the source episode, and participants may infer target hypotheses. Practice effects, regression, instrumentation change, and repeated testing can

mimic transfer. Latent carriers such as strategy and schema are model-dependent and should not be inferred from one trace.

The framework may privilege well-bounded computational tasks. Open-ended work, collective practice, and changing institutions can transform the target itself. In such cases transfer may be better described as recontextualization or development of activity rather than transport of stable content.

15. Reusable Artifacts

A. Transfer Ontology

Entities: source interaction, target interaction, agent or collective, experience-state component, carrier, cue, correspondence, adaptation, outcome, artifact, context, delay, measurement model.

Relations: acquired-in, retained-until, cued-by, corresponds-to, reused-in, adapted-for, mediated-by, expressed-as, improves, impairs, persists, decays, and is-measured-by.

B. Transfer Contract

Identity: source and target episode identifiers, boundaries, timescales, participants, and tasks.

Source learning: experience component changed, evidence, magnitude, and uncertainty.

Target shift: changed and preserved dimensions across semantic, structural, representational, opportunity, modality, device, social, and temporal axes.

Carrier: knowledge, procedure, strategy, belief, value, habit, attunement, artifact, or coordination practice.

Mechanism: retrieval, analogy, shared production, policy reuse, cueing, abstraction, compilation, or participation transformation.

Outcome: direct measures, derived metrics, latent estimates, direction, durability, and heterogeneity.

Counterfactual: control condition and assumptions supporting source attribution.

Failure conditions: no source learning, no retention, access failure, mismatch, negative transfer, confounding, or measurement non-invariance.

C. Experience Model

Experience is represented as a versioned update to distributed state rather than as a scalar amount. Each update records source episode, affected component, direction, confidence, persistence estimate, contexts in which it has been expressed, and known failure conditions.

D. Metric Registry

Metric	Numerator / estimand	Required controls	Prohibited interpretation
Target transfer effect	Counterfactual target difference	Baseline, prior exposure, target opportunity	Does not identify carrier
Savings	Trials/time saved to criterion	Criterion equivalence, practice dose	Not generality by itself
Generalization gradient	Effect across distance vector	Stable measurement and target difficulty	Distance is not one-dimensional
Strategy reuse	Posterior probability or coded recurrence	Trace comparability, alternative strategies	Not directly observed intention
Negative-transfer cost	Excess harm relative to control	Target semantics and feedback	Not evidence of low ability
Durability	Effect over delay	Retention interval and intervening experience	Immediate effect is not durable transfer

16. Conclusion

Existing theories of learning, transfer, expertise, adaptation, analogy, skill, reinforcement learning, and situated practice are sufficient to explain the known constituents of interaction transfer. The evidence does not justify a new primitive or a container called interaction memory.

A cross-disciplinary synthesis is nevertheless warranted. Computational interaction research requires a typed account because representation, opportunity, semantic structure, motor procedure, strategy, belief, artifact support, and

coordination can transfer differently. Transfer is best defined as a source-dependent difference expressed in a distinct target context under declared source-target relations and a valid counterfactual. The scientific burden is not to show that experienced agents perform differently, but to specify what changed, how it was accessed, why it applied, what was adapted, and which alternative explanations were excluded.

17. References

- Thorndike, E. L., & Woodworth, R. S. (1901). The influence of improvement in one mental function upon the efficiency of other functions. *Psychological Review*.
- Judd, C. H. (1908). The relation of special training to general intelligence. *Educational Review*.
- Singley, M. K., & Anderson, J. R. (1989). *The Transfer of Cognitive Skill*. Harvard University Press.
- Barnett, S. M., & Ceci, S. J. (2002). When and where do we apply what we learn? A taxonomy for far transfer. *Psychological Bulletin*, 128(4), 612-637.
- Bransford, J. D., Brown, A. L., & Cocking, R. R. (Eds.). (2000). *How People Learn*. National Academies Press.
- National Academies of Sciences, Engineering, and Medicine. (2018). *How People Learn II*. National Academies Press.
- Gentner, D. (1983). Structure-mapping: A theoretical framework for analogy. *Cognitive Science*, 7, 155-170.
- Gick, M. L., & Holyoak, K. J. (1983). Schema induction and analogical transfer. *Cognitive Psychology*, 15, 1-38.
- Anderson, J. R. (1982). Acquisition of cognitive skill. *Psychological Review*, 89, 369-406.
- Fitts, P. M., & Posner, M. I. (1967). *Human Performance*. Brooks/Cole.
- Ericsson, K. A., Krampe, R. T., & Tesch-Romer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100, 363-406.
- Lave, J. (1988). *Cognition in Practice*. Cambridge University Press.
- Greeno, J. G., Smith, D. R., & Moore, J. L. (1993). Transfer of situated learning. In *Transfer on Trial*.
- Beach, K. (1999). Consequential transitions: A sociocultural expedition beyond transfer in education. *Review of Research in Education*, 24, 101-139.
- Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.
- Engestrom, Y. (1987). *Learning by Expanding*. Orienta-Konsultit.
- Sutton, R. S., Precup, D., & Singh, S. (1999). Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112, 181-211.
- Taylor, M. E., & Stone, P. (2009). Transfer learning for reinforcement learning domains: A survey. *Journal of Machine Learning Research*, 10, 1633-1685.
- Pan, S. J., & Yang, Q. (2010). A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10), 1345-1359.
- Thorndike, R. L. (1949). *Personnel Selection: Test and Measurement Techniques*. Wiley.
- Norman, D. A. (2013). *The Design of Everyday Things*. Basic Books.
- Card, S. K., Moran, T. P., & Newell, A. (1983). *The Psychology of Human-Computer Interaction*. Erlbaum.
- Suchman, L. A. (1987). *Plans and Situated Actions*. Cambridge University Press.